Shehbaz Pathan

AI/ML Engineer | Agents, LLM evaluation, scientific ML

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SUMMARY

AI engineer working on agent evaluation and LLM inference. Built open source benchmarks for speculative decoding on multi step agent traces (spectrace) and for physics ML models (Coticula, PINNForge), and ship a production natural language to robot protocol agent at Opentrons. Research on trajectory level evaluation of enterprise workflow agents.

EXPERIENCE

AI Engineer, Opentrons Labworks Inc. | New York, NY — Aug 2026 to Present

- Shipped the Phase 0/1 MVP of an agent that turns scientists' plain English requests into runnable Opentrons lab robot protocols (OpenRouter LLM, FastAPI backend, chat UI).
- Build agentic AI, LLM infrastructure, and RAG pipelines for lab automation in Python, PyTorch, and LangChain/LangGraph.
- Evaluate agent reliability by testing protocols against scientist requests, catching failures before they reach the robot.

AI Automation Engineer Intern, Rocket Autoparts | Albuquerque, NM (Remote) — Feb 2026 to May 2026

- Cut order error rates 15% across tracked SKUs with ML forecasting and anomaly detection on fulfillment bottlenecks.
- Replaced manual handoffs in 3 core supply chain, inventory, and logistics workflows with LLM automation, including parsing unstructured emails and order forms into structured records.

Data Analyst, Automation Intern, BTS | New York, NY — Jun 2025 to Aug 2025

- Cut report turnaround from 2 days to under 4 hours (about 40% less manual processing) by automating Salesforce reporting and CRM ingestion, deduplication, and enrichment in Python for 5+ client accounts.
- Built Power BI dashboards on 15+ KPIs, adopted as the permanent leadership reporting layer for 3 consulting engagements.

Data Analyst, Dreamcare Developers Pvt. Ltd. | Pune, India — Jun 2022 to Jan 2024

- Guided roadmap priorities over 3 product cycles with regression and significance tests on 500K+ row datasets for 10+ clients; saved about 8 analyst hours per week by automating reporting.

PROJECTS

pera.ai | Python, Flask, SQLite, Docker | private repository

- Built a multi agent research terminal for private credit desks: an orchestrator routes to 5 specialist agents, and a deterministic rule based critic scores grounding, timeliness, tool reliability, and citations.
- Typed tools compute vintage, CDR, and concentration so the LLM never invents a figure; every result is labelled real, synthetic, derived, or customer. About 98% test coverage with pytest and GitHub Actions.

Coticula | PyTorch, NumPy, SciPy | github.com/shehbaz0101/coticula

- Showed FNO can beat PINO on L2 error while trailing it on PDE residual, using a four exam benchmark (Predict, Conserve, Counterfactual, Explain) on 1D Burgers and 2D heat.
- Released v0.2.0 with a 48 item rule graded exam and fail closed out of distribution and untrusted input flags.

PINNForge | PyTorch, FastAPI, Pydantic | github.com/shehbaz0101/pinnforge

- Recovered Burgers equation viscosity from 32 sparse sensors more accurately than a closed form estimator, which failed 9 of 30 hard out of distribution cases, using preregistered five seed Fourier neural operator studies.
- Built a PINN sandbox for harmonic, Burgers, and Poisson equations with a config driven CLI and rate limited API.

spectrace | Python, pytest, GitHub Actions | github.com/shehbaz0101/spectrace

- Built an open source benchmark that measures speculative decoding on multi step agent traces instead of chat data, reporting latency, draft accept rate, prefill token growth, and cost per successful task.
- Designed a two stage acceptance check: the LLM drafts each step and a rule based step grader scores tool success, progress, and abort; a live run accepted 22/22 steps across 5 traces. Free offline mock mode and keyless CI.

SmartBite | FastAPI, XGBoost, LightGBM | NYC health inspection risk platform

- Reached 87% precision across 5 violation categories at under 200ms latency.

SKILLS

Agents and Evaluation	Agentic AI, LLM and agent evaluation, trajectory level grading, speculative decoding, RAG, vector search, reranking, fine tuning, LangGraph, LangChain
Machine Learning	PyTorch, Hugging Face Transformers, FSDP/DDP, Fourier Neural Operators (FNO/PINO), Physics Informed Neural Networks (PINNs), TensorFlow, scikit-learn, XGBoost, LightGBM
Engineering	Python, SQL, TypeScript, R, FastAPI, Pydantic, pytest, GitHub Actions, CI/CD, AWS (S3, SageMaker), MLflow, PostgreSQL, ChromaDB, MongoDB
Analytics	Statistical modeling, significance testing, A/B testing, time series forecasting, Tableau, Power BI

EDUCATION

M.S. in Data Analytics and Visualization, Yeshiva University | New York, NY — May 2026

GPA 4.0/4.0 | Coursework: Machine Learning, Deep Learning, NLP, Time Series, Data Engineering

B.S. in Computer Applications, BAMU University | Pune, India | GPA 3.8/4.0 — May 2022